

Zerynth 2.0.5: The IoT Platform

The Internet of Things (IoT) gives users access to geographically distant places through smart applications running on it. Running these smart applications requires a platform to interface the software with embedded boards. Zerynth Studio is one such powerful integrated development environment (IDE) for embedded programming in Python that enables the IoT.

Go smart with Zerynth

Zerynth can be used for most of the popular prototyping boards. New hardware boards have been released for Zerynth Studio. As per the application, the Python programming is done in the prototype. This information is sent to the cloud or a database, from where it is controlled and distributed. Once the firmware is designed, it is uploaded for mass programming to suit industrial application. Then, users can operate it smartly using the IoT technology.

Zerynth IoT, initially known as Viper IoT, is now tailored for Zerynth 2.0.5. Imagine a user entering his home and without opening the refrigerator, he comes to know about the temperature and condition of food items inside the fridge. The user can also change the temperature of the fridge just through a Zerynth-powered application, called Refrigeration on Internet (ROI), running on his smartphone.

Not only the refrigeration system but also smart lights and their colour can be controlled using an app on smartphones. Zerynth has also come up with the latest LoRa WAN among many more applications.

Updates

The Zerynth stack includes tools as per applications, including Zerynth

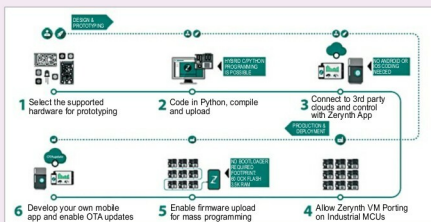


Fig. 1: The Zerynth workflow (Image courtesy: www.zerynth.com)



Fig. 2: Refrigeration of things (Image courtesy: <https://twitter.com/zerynth>)

virtual machine, Zerynth Studio, Zerynth advanced device manager and Zerynth app.

Zerynth Studio is an IDE that runs on Windows, MAC as well as Linux. Using Zerynth Studio programs can be managed through Python 3 or Hybrid C/Python. The user can select the RTOS as per the application in Zerynth Studio.

Zerynth Studio PRO, the professional version of Zerynth Studio, enables firmware over-the-air (FOTA) updates, so it becomes

easy to keep your solution updated with customer requirements. As updates are made, the rest of the firmware keeps running normally. So the offline time of the device is considerably reduced. The FOTA process can be performed manually by using the components of Zerynth Studio library.

Zerynth 2.0.5 has a lot of functionalities like advanced device manager packed together in the form of an upgrade to match industry 4.0 applications. It also